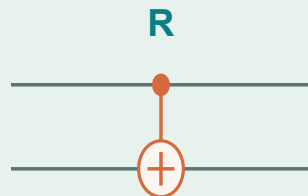
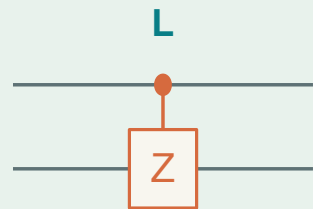
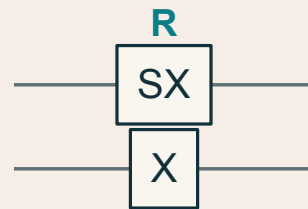
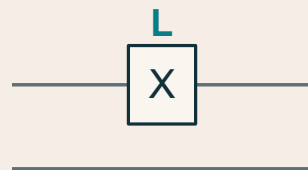


4 Compare spectra, including multiplicities

$$(L, R) = (CZ, CX)$$



$$(L, R) = (X \otimes I, SX \otimes X)$$





Eigenvalue test: does a unitary exist?

$$\text{eig}(L) = \{1, 1, 1, -1\}$$

$$\text{eig}(R) = \{1, 1, 1, -1\}$$

$$\text{eig}(L) = \{1, -1, 1, -1\}$$

$$\text{eig}(R) = \{1, -1, i, -i\}$$

$\text{eig}(L) = \text{eig}(R)$

$\text{eig}(L) \neq \text{eig}(R)$

✓ unitary S exists

✗ no unitary S exists

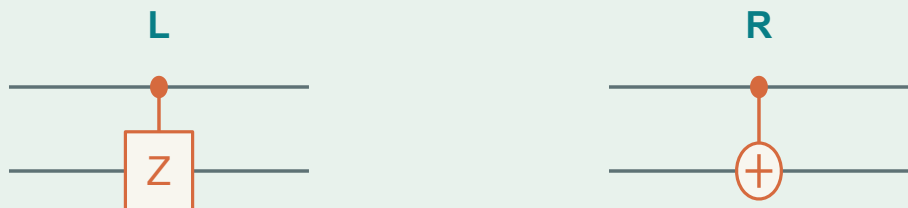
Keep $CZ \leftrightarrow CX$



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$$\text{eig}(L) \neq \text{eig}(R)$$

✗ no unitary S exists

Keep $CZ \leftrightarrow CX$

Solve for S

RUNNING EXAMPLE · L = CZ, R = CX

1 EXPAND

$$\begin{bmatrix} \text{[[S]} & \text{[[CZ]} \end{bmatrix} = \begin{bmatrix} \text{[[CX]} & \text{[[S]} \end{bmatrix}$$

$$\begin{bmatrix} s_{00} & s_{01} & s_{02} & -s_{03} \\ s_{10} & s_{11} & s_{12} & -s_{13} \\ s_{20} & s_{21} & s_{22} & -s_{23} \\ s_{30} & s_{31} & s_{32} & -s_{33} \end{bmatrix} = \begin{bmatrix} s_{00} & s_{01} & s_{02} & s_{03} \\ s_{10} & s_{11} & s_{12} & s_{13} \\ s_{30} & s_{31} & s_{32} & s_{33} \\ s_{20} & s_{21} & s_{22} & s_{23} \end{bmatrix}$$

Negate $|11\rangle$ column

Swap $|10\rangle / |11\rangle$ rows

2 EQUATE ENTRIES

$$\begin{aligned} s_{03} &= s_{13} = 0 \\ s_{20} &= s_{30} \\ s_{21} &= s_{31} \\ s_{22} &= s_{32} \\ s_{33} &= -s_{23} \end{aligned}$$

3 SYMBOLIC MATRIX

S =

$$\begin{bmatrix} a & b & c & 0 \\ d & e & f & 0 \\ g & h & j & k \\ g & h & j & -k \end{bmatrix}$$

10-dimensional linear space

Unitary instances: $S^\dagger S = I$